Arista Access Node HSR-ACC-27 Hardware/Environmental Fault Troubleshooting and Resolution SOP

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_2441

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve hardware/environmental faults on Arista Access Node HSR-ACC-27, which is experiencing ASIC memory parity errors and line card crashes due to power supply redundancy loss. The Arista Access Node HSR-ACC-27 is a crucial component of the network infrastructure at the HSR Layout Access site, providing access to the network for various endpoints and services. The hardware/environmental fault is expected to be caused by a combination of factors, including hardware malfunctions, environmental stressors, and power supply redundancy issues. The Arista EOS software, running on the Access Node, provides a robust set of features and tools to troubleshoot and resolve such issues. The SOP outlined below provides a step-by-step guide on how to identify the root cause of the issue, diagnose the problem, and resolve it using a combination of Arista EOS commands and external tools.

Pre-requisite Checks

- Ensure that the remote hands are dispatched to the HSR Layout Access site to physically inspect the Access Node and its surrounding environment.

- Verify the status of the redundant links connected to the Access Node to ensure that they are functioning correctly.

- Gather the necessary diagnostic tools and equipment to perform the troubleshooting steps outlined in this SOP.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware malfunction	High
Line card crashes due to power supply redundancy loss	Medium
Environmental stressors such as high temperatures or humidity	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource memory, Expected Output: Memory usage and allocation information for the Access Node, Description: This command provides a detailed view of the memory usage and allocation on the Access Node, which can help identify potential memory-related issues causing the ASIC memory parity errors.
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```
Command: show interface, Expected Output: Interface status and configuration information, Description: This command provides a detailed view of the interface status and configuration on the Access Node, which can help identify potential issues with the line card crashes.

Command: show power redundancy, Expected Output: Power supply redundancy status and configuration information, Description: This command provides a detailed view of the power supply redundancy status and configuration on the Access Node, which can help identify potential issues with the power supply redundancy loss.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: configure replace asic , Description: This command replaces the faulty ASIC with a new one, which can resolve the ASIC memory parity errors.

Step 2: Step: Replace the faulty line card, Command: configure replace linecard , Description: This command replaces the faulty line card with a new one, which can resolve the line card crashes.

Step 3: Step: Configure power supply redundancy, Command: configure power redundancy enable, Description: This command enables power supply redundancy on the Access Node, which can prevent power supply redundancy loss issues.

Post-Check Verification

```
Command: show system resource memory, Expected Output: Memory usage and allocation information for the Access Node, Description: This command verifies that the memory usage and allocation on the Access Node are within normal limits after the resolution steps.

Command: show interface, Expected Output: Interface status and configuration information, Description: This command verifies that the interface status and configuration on the Access Node are normal after the resolution steps.

Command: show power redundancy, Expected Output: Power supply redundancy status and configuration information, Description: This command verifies that the power supply redundancy status and configuration on the Access Node are normal after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original ASIC configuration, Command: configure default asic , Description: This command restores the original ASIC configuration, which can resolve any issues caused by the replacement of the ASIC.

- Step: Restore the original line card configuration, Command: configure default linecard , Description: This command restores the original line card configuration, which can resolve any issues caused by the replacement of the line card.

- Step: Disable power supply redundancy, Command: configure power redundancy disable, Description: This command disables power supply redundancy on the Access Node, which can resolve any issues caused by the configuration of power supply redundancy.